

Teaching Exam

**Subject –
Computer
Science**

**COMPUTATIONAL THINKING
AND PROGRAMMING**

Lecture No –4



By-Satyendra Chaudhary Sir

Topics to be Covered



Topic

Computational Thinking and Programming

S

Strings - (slicing)

List -

Tuples -

is is not $\rightarrow$ Identity

in not in $\rightarrow$ membership

$a \text{ is not } b \rightarrow \underline{\text{False}}$

a in S $\Rightarrow$ True / False

a is b $\rightarrow$ True / False

a == b

a is not b $\rightarrow$ False

Type conversion $\begin{cases} \rightarrow \text{automatically} \\ \rightarrow \text{type casting} \end{cases}$

$a = \text{input}(\text{"Enter the value of a: "})$

$a = \text{int}(\text{input}(\text{"Enter the value of a: "}))$

Unary operators

+

-

$\sim$

$a = +5$

$a = -5$

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

$$\sim(A \cup B) = \sim A \cap \sim B$$

Boolean

$$\overline{(A \cup B)} = \bar{A} \cap \bar{B}$$

$$\sim(A \cup B) = \sim A \cap \sim B$$

strings →

strings slicing →

a = I - study on Physics wallah.

0 1 2 3 4 5 6 7 8 9

-4 -3 -2 -1

a [- : _ :]

a [- : _ :]

starting assign ending assign + 1

gap (skip)

-7 -6 -5 -4 -3 -2

-1

a[0]

0 1 2 3 4 5 6

wallah.

$$a[-1:-5] = \text{ , } \swarrow$$

$$a[-5:-1] = \underline{\text{allow}^x}$$

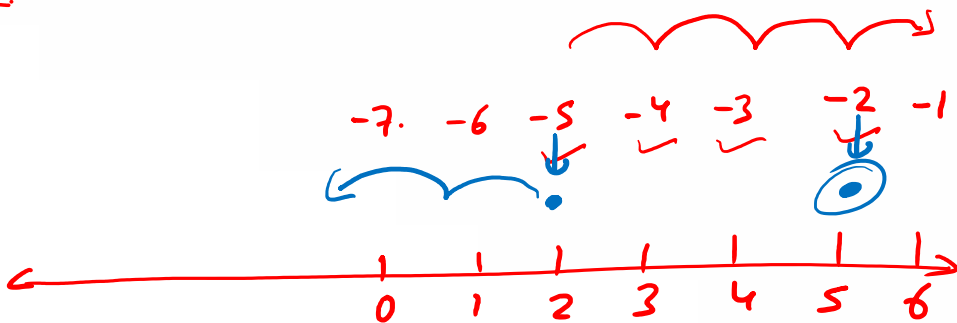
$$a[\text{---}:\text{---}:\text{---}]$$

gap

$$a[-1:-5:-1]$$

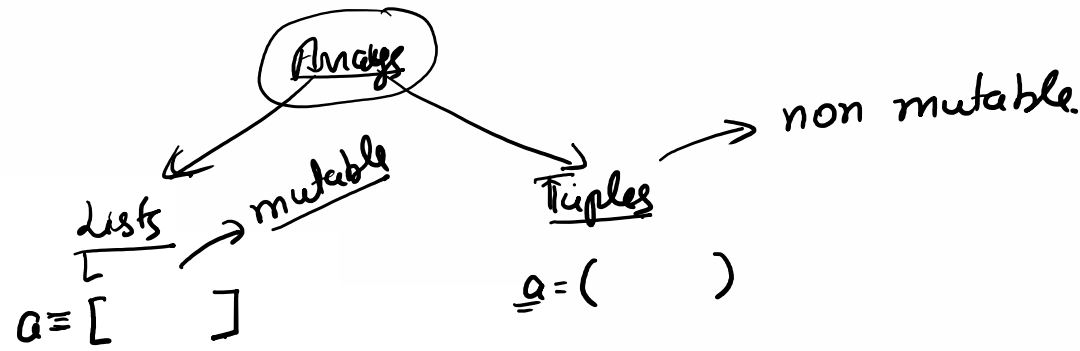
gap

$$-1+1=0$$



$$a[-5:-2:-1]$$

- ① str.endswith(u) # True / False
a.endswith(h) # True
- ② str.capitalize() # capitalize.
- ③ str.find(a) # a at first occurrence.
↓
'chai'

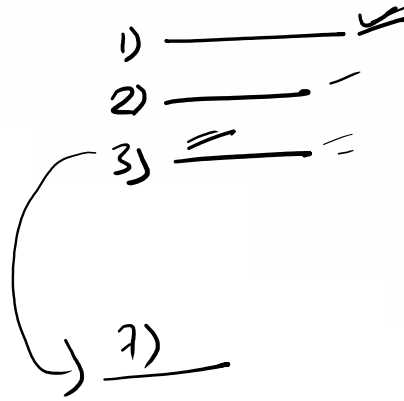


Dictionary → non mutable
 $a = \{$

$-4 \quad -3 \quad -2 \quad -1$
 $a = [3, 'five', 12, 2+3j]$
 $\uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 $0 \quad 1 \quad 2 \quad 3$

$a = (3, 'five', 12, 2+3j)$

flow of statements



list: append. (2+3j)

D
Else -
import()

DSA → C++, Java

Pauls → Python, C++, Java



2 mins Summary

Topic

Computational Thinking and Programming

THANK YOU